Jay Pokale

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.56
Sri Chaitanya Junior Kalashala Dulapally	Class XII (Telangana State Board)	2018	92.60%
Muncipal High School Kalmeshwar	Class X (Maharashtra State Board)	2016	85.40%

SCHOLASTIC ACHIEVEMENTS

LeetCode Guardian: Peak Rating 2285; solved 1,637 problems, including 341 Hard, across 44 contests.

Codeforces Expert: Rating 1799 (**rdx_panther**).

Founder, Dare2Solve: Reached 20,000+ followers; built a math problem-solving community from scratch since 2021, publishing 209 puzzles with worked solutions, including 181 across two consecutive years.

OPEN SOURCE

Chisle: Token-Efficiency Toolkit for AI Coding Agents | 600+ GitHub stars · 35 forks

- JavaScript · Node.js · Claude Code PluginReduced AI-agent output-token costs by 48% across 20 live benchmark tasks, achieving the lowest worst-case overhead among evaluated tools.
- Built a deterministic output-compression hook that saved ~46% tokens across 171 real sessions.
- Created one-command installation for 11 AI coding agents, including Claude Code, Cursor, Codex, Gemini, and GitHub Copilot, with zero dependencies.
- Published complete benchmark results, including unsuccessful runs, for reprehensibility.

Ballerina (WSO2): Open-Source Contributor

- React · Next.jsMerged 4 pull requests into WSO2's Ballerina programming-language website.
- Fixed layout shifts on the downloads page and button hover states, repaired a broken Windows certificate link, and refactored release-notes URL handling.

typed-numarray: High-Performance Growable Typed Arrays

- JavaScript · npm · Jest · GitHub ActionsBuilt growable typed arrays supporting 10 numeric types, including 64-bit BigInt arrays.
- Implemented constant-time shift/unshift operations for 100,000 shifts on standard arrays (~280× faster).
- Implemented counting sort for bounded integers, achieving 17× speedup over Array.sort.
- Published and maintained the npm package since 2023 with automated testing and releases.

competitive: Data Structures Library for Competitive Programming

- JavaScript · AlgorithmsBuilt a JavaScript competitive-programming library providing heaps, segment trees, Fenwick trees, union-find, Dijkstra, KMP, Z-function, and Manacher's algorithm.
- Implemented safe modular arithmetic without BigInt to avoid JavaScript's 53-bit integer limitation.
- Used the library across 1,142 LeetCode solutions in JavaScript while reaching LeetCode Guardian (top 1% worldwide).

PROJECTS

GST Legal RAG: Production AI Legal Assistant for Indian Tax Law

- Python · RAG · Hybrid Retrieval · LLM Evaluation · CIBuilt a production retrieval system for GST law that provides

cited answers and abstains when retrieval confidence is insufficient.
- Separated tax arithmetic from legal retrieval using an exact calculator, ensuring numerical answers are computed rather than predicted.
- Designed an LLM-as-judge evaluation harness that rejects baseline results when the judge model is unhealthy.
- Built crash-safe document ingestion with resumable processing, allowing failures at document 80,000 to resume from 80,001.

AuthorsLog: Multi-Author Publishing Platform

- SolidStart · tRPC · Zod · MongoDB · JWT · SendGrid · Next.js · ExpressBuilt a two-app publishing platform with a public reader site and separate writer dashboard, sharing a type-safe tRPC + Zod API over MongoDB.
- Implemented secure authentication with bcrypt password hashing, JWT sessions, and SendGrid email verification.
- Built a rich-text editor with image uploads, draft/published workflows, author profiles, and post statistics.
- Added SEO features including readable URLs, dynamic sitemap and robots.txt; later rebuilt the platform in Next.js with a block-based editor and standalone Express backend.

Office Management Platform: Cross-Platform Operations App

- React Native (Expo) · Express · MongoDB · Clerk · CloudinaryShipped a single-codebase operations platform for Android, iOS, and Web, tracking materials, services, purchases, and payments.
- Built a hardened REST API using Helmet, compression, and rate limiting at 100 requests per 15 minutes per IP.
- Uploaded device photos directly to Cloudinary, keeping image data off the application server.
- Implemented count-free pagination by fetching one extra record to detect subsequent pages.

jaypokale.me: Personal Site & Technical Blog

- Next.js 16 · TypeScript · Three.js / GLSL · GitHub ActionsBuilt a WebGL particle portrait with ~12,000 points rendered through a custom GLSL shader.
- Integrated live GitHub, LeetCode, and Codeforces statistics with hourly refreshes and fallback handling.
- Implemented structured data, automatic article link-preview images, and instant Bing indexing notifications.
- Built a Markdown blog with CI checks and Mermaid diagrams pre-rendered into light/dark theme-aware images.

SKILLS

Languages - Fluent in semicolons: C++ | TypeScript | JavaScript | Python | Go

Algorithms - Data Structures | Graph Theory | Dynamic Programming | Number Theory | Computational Geometry

AI / Retrieval - RAG Pipelines | Hybrid Retrieval (BM25 + Dense) | Cross-Encoder Reranking | LLM Evals & Judges | Multi-Agent Systems | MCP | PyTorch | NLP

Web - React / Next | SolidJS | Node / NestJS | tRPC | GraphQL | Tailwind | three.js | GSAP / Motion

Infra & Security - MongoDB | MySQL | PostgreSQL | Docker | GitHub Actions | Cloudflare Workers | Vercel | Web Security / CTFs | Network Security | Arch Linux (btw)

POSITIONS OF RESPONSIBILITY

Class Representative (CR) IIT Hyderabad CSE

Founding Member, Calisthenics Club IIT Hyderabad

Hostel Representative Ramanuja - IIT Hyderabad

Teaching Assistant (TA) - Data Structures and Algorithms (ID 2230) IIT Hyderabad

Teaching Assistant (TA) - Theory of Computation (CS2030) IIT Hyderabad

Teaching Assistant (TA) - Data Mining (CS5600) IIT Hyderabad

EXTRACURRICULAR

Powerlifting: Won 14 medals in powerlifting competitions.

Weightlifting: Won 3 medals in weightlifting competitions.

Mathematics Mentor, Dare2Solve: Guided students and community members in solving challenging mathematics problems and developing problem-solving skills.

Backflip Coach, IIT Hyderabad: Conduct weekly backflip training sessions at IIT Hyderabad every week.